

SEAN MACBRIDE

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RESEARCH INTERESTS

- Experimental and Observational Cosmology
- Multi-Messenger Astrophysics
- Astronomical Instrumentation Development
- Systems Design for Astronomy

EDUCATION

Ph.D. in Physics, University of Zürich May 2024 - 2027

Thesis title: *Standard Siren Cosmology: From Instrumentation to Inference*

Advisor: Professor Marcelle Soares-Santos

M.Sc. in Physics, University of Michigan-Ann Arbor Aug 2022 - May 2024

Advisor: Professor Marcelle Soares-Santos

B.A. in Physics with Honors, Wheaton College, Massachusetts Aug 2016 - May 2020

Thesis title: *Characterizing the Dust and Cold-Gas Content of Nearby Star-Forming Galaxies.*

Advisor: Professor Dipankar Maitra

SCIENTIFIC COLLABORATIONS

Rubin Observatory Legacy Survey of Space and Time (LSST) 2023 - Present

I hold roles in both the Rubin Project and the LSST Dark Energy Science Collaboration (DESC).

Rubin Observatory Project Team: Led LSST-Camera (LSSTCam) focal plane commissioning to Rubin Observatory in Cerro-Pachón in Chile, from receipt on summit through early science operations, including testing of the focal plane in-laboratory and on-telescope. Characterized defects in the LSSTCam focal plane CCDs. Developed custom image processing tool to monitor the stability of defects in the LSSTCam focal plane. Led the Target-of-Opportunity (ToO) observing program during Rubin early operations.

LSST DESC: Forecasted the constraining power of dark sirens using simulated LSST data under different conditions. Leading a dark-siren analysis using LSST commissioning data (Data-Preview 2) and public data from the International Gravitational Wave Network (IGWN), the first dark siren detection using LSST data.

International Gravitational-Wave Observatory Network (IGWN) 2025 - Present

Led dark siren analysis using DECam Local Volume Exploration survey (DELVE) survey data for IGWN O4c data, including galaxy catalog preparation and systematic-uncertainty quantification. Coordinated time-critical ToO observations with IGWN researchers, interfacing with Rubin Observatory alert brokers to rapidly disseminate information about gravitational wave (GW) candidates. Contributed to dark siren analysis using Dark Energy Survey (DES) data for IGWN O4b data.

Dark Energy Survey (DES) 2023 - Present

Performed observations, image processing, and analysis of images from Dark Energy Camera (DECam) observations in response to GW triggers from the International Gravitational Wave Network (IGWN) observing run 4 (O4) for the Dark Energy Survey gravitational wave (DESGW) observing program. Implemented improvements to DESGW alert and image processing pipelines, including improvements to the telescope strategy and real-time alert monitoring. Led collaboration between the Japanese collaboration for Gravitational wave ElectroMagnetic follow-up (J-GEM) and DESGW through joint observations of GW event S250328ae using DECam and the Prime Focus Spectrograph (PFS) at Subaru Observatory.

Robotic Positioners for Future Spectroscopic Surveys

2023 - Present

Tested Dark Energy Spectroscopic Instrument (DESI) robotic positioners to increase movement precision and improve low performing robotic positioners on the DESI focal plane. Characterized a common failure mode of DESI positioners and resolved the failure with a reproducible solution. Authored and submitted an internal report and presented the results of this study to the DESI focal plane working group. Led dynamic testing of prototype robotic positioner performance through lifetime, focal-ratio degradation, and fiber angle tests using different telescope configurations. Tested prototype robotic positioners for Stage-V surveys, focusing on characterization of tilt induced by different orientations.

Large Lenslet Array Magellan Spectrograph (LLAMAS)

2020 - 2022

Assembled and modified opto-mechanical mounts and ground support equipment for the LLAMAS instrument, including verification of mechanical tolerance for custom-fabricated parts using a coordinate measurement machine. Verified spectrograph grating efficiency through optical tests, and integrated the spectrograph camera lenses into their bezels for the 24 camera spectrograph. Integrated the 2,400 AR-coated fibers to the spectrograph by bonding with optical adhesive, with 100% accuracy and zero fiber loss.

PROFESSIONAL APPOINTMENTS, SERVICE, AND LEADERSHIP

2026 - 2028	LSST DESC Speakers Bureau Co-Chair
2026 - 2028	LSST DESC Collaboration Council Member
2025 - 2027	Rubin Observatory ToO Observer
2024 - 2027	LSSTCam Summit Support Scientist
2022 - 2024	Elected member, University of Michigan Physics Graduate Council
2022 - 2024	Member, University of Michigan Physics DEI committee
2022 - 2024	Member, University of Michigan APS chapter
2016 - 2020	Member and peer advisor for the Wheaton College physics club

TEACHING

2026	Graduate Instructor, Introduction to Scientific Computing, University of Zürich
2025	Graduate Instructor, Introduction to Astroparticle Physics, University of Zürich
2023-2024	Lead Graduate Instructor, Introductory Physics for Life Sciences, University of Michigan-Ann Arbor
2022-2023	Graduate Instructor, Introductory Physics for Life Sciences, University of Michigan-Ann Arbor
2018-2019	Undergraduate Teaching Assistant, Introductory Physics, Wheaton College, MA

CONFERENCE ORGANIZATION

2025	EAS Annual Meeting Special Session 27: The Rubin Observatory Legacy Survey of Space and Time: from Data Previews to Full Survey Operations
2024	Conferences for Undergraduate Women in Physics (CUWiP) 2024
2023	DESI Collaboration Meeting, LOC
2023	University of Michigan-Ann Arbor Physics Graduate Student Symposium

OUTREACH

- 2023-2024 Led group of high school students in Burlington VT to observe the 2023 annular eclipse and 2024 total eclipse. Prepared and delivered lectures for students about the physics of the sun, solar observations, and eclipses through virtual and in person presentations.
- 2023 Designed art installation in University of Michigan Physics help room, highlighting struggles overcome by recent graduates of the physics program and success in their later lives.
- 2016-2020 Showcased features of different telescopes to local tour groups at Wheaton College Observatory. Operate and adjust telescopes to show appropriate stars, objects, and events based on sky conditions. Educated children and adults in how to locate objects and explained features about the objects on sky.

MENTORSHIP

- 2026 Alexis Gaute: Summer Masters Student at University of Zürich.
- 2025-2026 Guandi Zhao: Masters Student at University of Zürich.
- 2024-2025 Banan Yamani: Masters Student at University of Zürich. Current Physics PhD student at Paul Scherrer Institute.
- 2024-2025 Yavuz Gencel: Masters Student at University of Zürich.
- 2023-2024 Elise Kesler: Undergraduate student at University of Michigan-Ann Arbor. Current physics PhD student at CU Boulder.
- 2023-2024 Tim Fanning: Undergraduate student at University of Michigan-Ann Arbor. Current nuclear physics PhD student at Indiana University Bloomington.

PRESS

- 08/2025 The Vera Rubin Observatory is ready to revolutionize astronomy - Science News.
- 06/2025 Mapping the Universe at the Rubin Observatory in Chile - BBC.
- 06/2025 World's Largest Digital Camera Snaps Its First Photos of the Universe - Wall Street Journal.
- 04/2024 NASA volunteer documents eclipse from Burlington to help study sun's atmosphere - Vermont Public Radio.

SELECTED PUBLICATIONS

Below are selected publications from different collaborations and projects. A full list of my publications can be found on my Google Scholar profile or on the astrophysics data system.

Astronomical Instrumentation

1. The On-Sky Performance of the LSST Camera CCD Array. **MacBride**, Roodman, ... 30 Jun 2026.
2. First year into operations of LLAMAS, the Large Lenslet Array Magellan Spectrograph: design, build and commissioning overview, results and performance, lessons learned. Furesz, ... **MacBride**, ... in prep.
3. Characterizing robotic positioners under the influence of changing gravity vectors for future spectroscopic surveys. Wüthrich, ... **MacBride**, ... 25 Jun 2026.

4. LSST Camera Electro-Optical Test (Run 7) Results. The LSST Camera group: Antilogus, ... **MacBride**, ... 5 May 2025.
5. The Stage-5 Spectroscopic Experiment. Besuner, Dey, ... **MacBride** ... 12 Mar 2025.

Observational and Experimental Cosmology

1. Measurement of the Hubble constant using the Dark Energy Survey Year 6 Gold galaxy catalog and the fourth Gravitational-Wave Transient Catalog. McMahon, ... **MacBride**, ... 5 Feb 2026.
2. Standard Sirens in 2040s: Probing the Cosmic Expansion History with Gravitational Waves and Spectroscopic Galaxy Surveys. Borghi, ... **MacBride**, ... 20 Dec 2025.

Multi-Messenger Astrophysics

1. A Joint Search for the Electromagnetic Counterpart to the Gravitational-Wave Binary Black-Hole Merger Candidate S250328ae with the Dark Energy Camera and the Prime Focus Spectrograph. Zhang, Kokubo, **MacBride**, ... 1 Aug 2025.
2. NSF-DOE Vera C. Rubin Observatory Observations of Interstellar Comet 3I/ATLAS (C/2025 N1) . Chandler, Bernardelli, ... **MacBride**, ... 21 July 2025.
3. Rubin ToO 2024: Envisioning the Vera C. Rubin Observatory LSST Target of Opportunity program. Andreoni, Margutti, ... **MacBride** ... 7 Nov 2024.

Astronomical Systems Development

1. The Rubin Observatory Target-of-Opportunity Program in the First Year of Operations. **MacBride**, Ribeiro, ... 30 Jun 2026.
2. LSSTCam installation and commissioning. Fanning, ... **MacBride**, ... in prep.
3. Scheduling the LSST: commissioning the Rubin Feature Based Scheduler. Jones, ... **MacBride**, ... in prep.

INVITED TALKS

07/2026	Plenary Talk, Rubin Community Workshop 2026, SLAC National Accelerator Laboratory
07/2026	Seminar, Dark Energy School, LSST DESC Collaboration Meeting, Paris, France
07/2026	Talk, SPIE Astronomical Telescopes + Instrumentation, Copenhagen, Denmark
05/2026	Talk, Rubin ToO 2026 Workshop
11/2025	Talk, Rubin-LSST France Meeting at LPSC Grenoble, France
09/2025	Talk, 5th Philip Wetton Workshop, Oxford University
07/2025	Talk, Rubin Community Workshop 2025, Tucson, AZ
06/2025	Seminar, Massachusetts Institute of Technology
02/2025	Plenary Talk, DESC Collaboration meeting
11/2021	Colloquium, Wheaton College, Norton MA

CONTRIBUTED TALKS

07/2026	Talk, Rubin Community Workshop 2026, SLAC National Accelerator Laboratory
07/2026	Talk, SPIE Astronomical Telescopes + Instrumentation, Copenhagen, Denmark
04/2026	Seminar, Stanford University
03/2026	Talk, LIGO-Virgo-KAGRA Collaboration Meeting, Pisa, Italy
03/2026	Talk, LIGO-Virgo-KAGRA Collaboration Meeting, Pisa, Italy
05/2025	Talk, The Dark Energy Survey Collaboration Meeting 2025,
03/2025	Talk, APS March meeting 2025, Anaheim CA, USA
05/2020	Talk, Wheaton College, MA
09/2018	Seminar, Wheaton College, Norton MA
08/2018	Talk, Rutgers University